

OvCWD Scenario Design and Analysis

CWD status	Focus	Goals	Questions	Scenarios	Metrics
Not yet detected: Pre-introduction phase or pre-establishment phase	Surveillance	Assess CWD status	CWD free?	Baseline harvest and CWD testing	Likelihood of undetected CWD
			Can we detect CWD promptly if introduced?	Current harvest and CWD testing; CWD introduction – dispersing yearling males/females/adult deer	Detection probability, time to first detection, true population prevalence at first detection
			Improve surveillance: targeted surveillance strategies. How many deer to sample? Which deer to sample? Proportion of harvest tested?	Compare and contrast alternate strategies for CWD surveillance	Detection probability, time to first detection, true population prevalence at first detection
	Enhanced surveillance	Potential routes of CWD introduction	Where to focus surveillance efforts? GIS data	Corridors: riparian corridors, migratory routes	Detection probability
		Hotspots	GIS data	Proximity to known CWD areas; captive cervid facilities	Detection probability
	Proactive management	Deer population management	Design optimum harvest strategies	Compare and contrast current and alternate harvest strategies	Population growth rate (λ); sex age-class wise harvest
		Resilient deer population	Prevent establishment when introduced	CWD trajectory under current and alternate harvest strategies	Demographic competence/outbreak probability/fadeout probability
		Public practices	Assess the role of public practices like baiting, supplemental feeding, attractants, carcass movement.	CWD trajectory	Force of infection, population prevalence

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CWD status	Focus	Goals	Questions	Scenarios	Metrics
Recently detected: Pre-establishment phase or transition phase or endemic phase	Surveillance	Assess the phase of the outbreak	Are there more cases of CWD around the first detection? Size of CWD management zone/surveillance zone? Core area size? Duration of focused surveillance?	Varying levels of local prevalence and evaluate surveillance strategies	Detection probability; likelihood of undetected CWD
	Management	Eliminate CWD; slow the spread and establishment	Harvest strategies	Local strategies: targeted culling - Core area for targeted culling, intensity of targeted culling Regional harvest strategies	Force of infection, prevalence, number of infected deer, CWD trajectories
Endemic	Management	Control the spread, slow population decline	Design appropriate harvest strategies	Age class sex wise harvest design	Force of infection, prevalence, lambda